Theorem 5.5.1: Let $r, s \in \mathbb{R}_+$ s.t. $p := \frac{s}{r} \notin \mathbb{Q}$. Then

$$D_{b_1,\dots,b_\ell} \cdot 1 = \sum_{\lambda} t^{a(\lambda)} q^{\mathrm{dinv}_p(\lambda)} \omega\left(\mathcal{G}_{\nu(\lambda)}(X;q^{-1})\right)$$

where
- the sum is over λ: lattice path from $(0,\lfloor s \rfloor)$ to $(\lfloor r \rfloor, 0)$ lying weakly below the line $y=-px+s$ through $(0,s)$ and $(r,0)$,
- $a(\lambda)$ = # lattice squares enclosed between λ and δ where δ is the highest path from $(0,\lfloor s \rfloor)$ to $(\lfloor r \rfloor, 0)$ weakly below $y=-px+s$.
- b_i = # S-steps in δ along $x=i-1$ $\quad (i=1,\dots,\ell=\lfloor r \rfloor+1)$,
- $\mathrm{dinv}_p(\lambda)$ defined in Def 5.4.1
- $\nu(\lambda) = \sigma(\beta/\alpha) = (\beta_{w_0\sigma(1)},\dots,\beta_{w_0\sigma(\ell)})/(\alpha_{w_0\sigma(1)},\dots,\alpha_{w_0\sigma(\ell)})$

Proof: We prove the equivalent identity

$$\omega(D_{b_1,\dots,b_\ell}\cdot 1) = \sum_{\lambda} t^{a(\lambda)} q^{\mathrm{dinv}_p(\lambda)} \mathcal{G}_{\nu(\lambda)}(X;q^{-1})$$

By Corollary 3.7.2 and Lemma 4.1.6, both $\omega(D_{b_1,\dots,b_\ell}\cdot 1)$ and $\mathcal{G}_{\nu(\lambda)}(X;q^{-1})$ involve only $s_\ell(X)$ with $\ell(\lambda) \le \ell$.
Hence it suffices to show the identity when evaluated in ℓ variables $x_1,\dots,x_\ell$.
By Corollary 3.7.2, $\omega(D_{b_1,\dots,b_\ell}\cdot 1) = (\mathcal{H}_{b_1,\dots,b_\ell})_{\mathrm{pol}}$. Hence we want to prove

$$(\mathcal{H}_{b_1,\dots,b_\ell}(x_1,\dots,x_\ell))_{\mathrm{pol}} = \sum_{\lambda} t^{a(\lambda)} q^{\mathrm{dinv}_p(\lambda)} \mathcal{G}_{\nu(\lambda)}(x_1,\dots,x_\ell;q^{-1}).$$

We now prove this using 5.3.1.

Define $b_i' := \lfloor s-p(i-1) \rfloor - \lfloor s-pi \rfloor$ and δ' be the highest SE lattice path under $y=-px+s$ starting at $(0,\lfloor s \rfloor)$ and extends forever. ($\therefore$ By Remark 5.3.2, b_i' = # S-steps of δ' on $x=i-1$)
Then δ' agrees with δ between $x=0$ and $x=\ell-2$ (inclusively) and $b_i' = b_i$ $\forall i \in \{1,\dots,\ell-1\}$.
Also $b_\ell \le b_\ell'$ because δ stops at $(\ell-1,0)$ while δ' may extend below the x-axis. (See §5.3)
By Theorem 5.3.1 with b_i' in place of b_i, set $u=0$, $v=b_\ell'-b_\ell$ and take pol on both sides, we get

$$\left(\mathcal{H}_{b_1',b_2',\dots,b_{\ell-1}',b_\ell'-(b_\ell'-b_\ell)}(x_1,\dots,x_\ell)\right)_{\mathrm{pol}} = \sum_{a_1,\dots,a_{\ell-1}\ge 0} t^{|\vec{a}|} \mathcal{L}^{\sigma}_{((b_\ell',\dots,b_1')+(-b_\ell'+b_\ell,a_{\ell-1},\dots,a_1))/(a_{\ell-1},\dots,a_1,0)}(x_1,\dots,x_\ell;q)$$

$$\left(\mathcal{H}_{b_1',b_2',\dots,b_{\ell-1}',b_\ell}(x_1,\dots,x_\ell)\right)_{\mathrm{pol}} = \sum_{a_1,\dots,a_{\ell-1}\ge 0} t^{|\vec{a}|} \mathcal{L}^{\sigma}_{((b_\ell',\dots,b_1')+(0,a_{\ell-1},\dots,a_1))/(a_{\ell-1},\dots,a_1,0)}(x_1,\dots,x_\ell;q)$$

$b_i' = b_i \ \forall 1 \le i \le \ell-1$

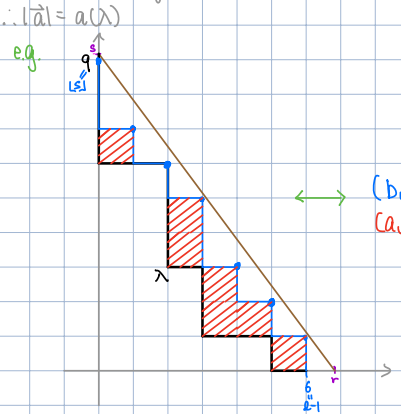
$$\left(\mathcal{H}_{b_1,b_2,\dots,b_{\ell-1},b_\ell}(x_1,\dots,x_\ell)\right)_{\mathrm{pol}} = \sum_{a_1,\dots,a_{\ell-1}\ge 0} t^{|\vec{a}|} \mathcal{L}^{\sigma}_{((b_\ell,b_{\ell-1},\dots,b_1)+(0,a_{\ell-1},\dots,a_1))/(a_{\ell-1},\dots,a_1,0)}(x_1,\dots,x_\ell;q)$$

By Corollary 4.5.7, $\mathcal{L}^{\sigma}_{((b_\ell',b_{\ell-1},\dots,b_1)+(0,a_{\ell-1},\dots,a_1))/(a_{\ell-1},\dots,a_1,0)}(x_1,\dots,x_\ell;q) \ne 0 \Rightarrow (a_{\ell-1},\dots,a_1,0) \overset{\text{coordinate-wise}}{\le} (b_\ell,\dots,b_1)+(0,a_{\ell-1},\dots,a_1)$

i.e. $a_{\ell-1} \le b_\ell$, $\quad a_i \le a_{i+1} + b_{i+1} \quad \forall\, 1 \le i \le \ell-2$ ——— $(*)$

These conditions are precisely for there to exist a unique lattice path λ from $(0,\lfloor s \rfloor)$ to $(\lfloor r \rfloor,0)$ s.t.
a_i = # lattice squares in the ith column (i.e. between $x=i-1$ to $x=i$) of the region between λ and δ.
$\therefore |\vec{a}| = a(\lambda)$

e.g.



$\longleftrightarrow$ $(b_1,b_2,b_3,b_4,b_5,b_6,b_7) = (2,1,1,2,1,1,1)$
$(a_1,a_2,a_3,a_4,a_5,a_6,a_7) = (1,0,2,2,1,1,0)$

Also, when $(*)$ holds, by Def. 5.4.2, the LLT data for λ is given by:
$\beta = (b_\ell,\dots,b_1)+(0,a_{\ell-1},\dots,a_1)$, $\alpha = (a_{\ell-1},\dots,a_1,0)$, $\sigma \in S_\ell$ s.t. $\sigma(1),\dots,\sigma(\ell)$ are in the same relative order as $c_\ell,\dots,c_1$; $c_i := \lfloor s-p(i-1) \rfloor$ (as in Theorem 5.3.1)

$\therefore$ By Corollary 4.5.7, with $(*)$ holds and the LLT data above, we have

$$\mathcal{L}^{\sigma}_{((b_\ell, b_{\ell-1},\dots,b_1)+(0,a_{\ell-1},\dots,a_1))/(a_{\ell-1},\dots,a_1,0)}(x_1,\dots,x_\ell;q)\,\mathrm{pol} = q^{h_\sigma(\beta/\alpha)}\,\mathcal{G}_{\nu(\beta/\alpha)}(x;q^{-1}) = q^{h_\sigma(\beta/\alpha)}\,\mathcal{G}_{\nu(\lambda)}(x;q^{-1})$$

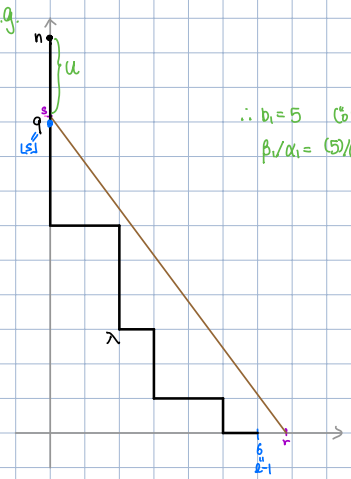
By Prop. 5.5.4, $h_\sigma(\beta/\alpha) = \mathrm{dinv}_p(\lambda)$,

$$\therefore \left(\mathcal{H}_{b_1,b_2,\dots,b_{\ell-1},b_\ell}(x_1,\dots,x_\ell)\right)_{\mathrm{pol}} = \sum_{a_1,\dots,a_{\ell-1}\geq 0} t^{|\vec{a}|}\,\mathcal{L}^{\sigma}_{((b_\ell,b_{\ell-1},\dots,b_1)+(0,a_{\ell-1},\dots,a_1))/(a_{\ell-1},\dots,a_1,0)}(x_1,\dots,x_\ell;q)$$

$$= \sum_{\lambda} t^{a(\lambda)}\, q^{\mathrm{dinv}_p(\lambda)}\,\mathcal{G}_{\nu(\lambda)}(x_1,\dots,x_\ell;q^{-1}) \quad \text{and result follows.} \qquad \square$$

Remark 5.5.2: If $u>0$, the proof still works for a more general version of Theorem 5.5.1 in which the sum is over lattice paths λ that starts at $(0,|s|+u)$, i.e. extending the south run on $x=0$. The only modifications are (i) b_1 counts the S-steps on $x=0$ of λ including the extension (ii) the row in $\nu(\lambda)$ corresponding to S-steps in λ on $x=0$ is also extended accordingly.

e.g.



$\therefore b_1 = 5$ ("old" $b_1 = 3$)

$\beta_1/\alpha_1 = (5)/(0)$ ("old" $\beta_1/\alpha_1 = (3)/(0)$)

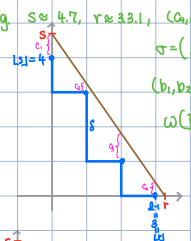
e.g. $s \approx 4.7$, $r \approx 3.3.1$, $(a_1,a_2,a_3,a_4) \approx (0.44, 0.86, 0.28, 0.70)$, $p \approx 1.42$

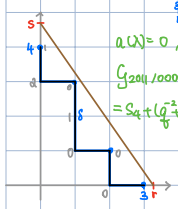
$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix}$

$(b_1,b_2,b_3,b_4) = (1,2,1,0)$

$$\omega(D_{1210}\cdot 1)(x_1,x_2,x_3,x_4) = \mathcal{H}_{(1,2,1,0)}(x_1,x_2,x_3,x_4)_{\mathrm{pol}} = \sigma\left(\frac{x_1x_2^2x_3\overbrace{(1-qt\frac{x_1}{x_3})(1-qt\frac{x_1}{x_4})(1-qt\frac{x_2}{x_4})}^{\prod_{1\leq i<j\leq 4}(1-qt\frac{x_i}{x_j})}}{\prod_{1\leq i<j\leq 4}(1-q\frac{x_i}{x_j})(1-t\frac{x_i}{x_j})}\right)_{\mathrm{pol}} = \text{sum of } t^{a(\lambda)}q^{\mathrm{dinv}_p(\lambda)}\mathcal{G}_{\nu(\lambda)}(x;q^{-1})$$

for the following 7 λ's.

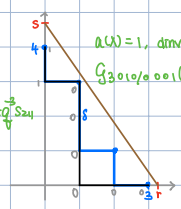




$a(\lambda)=0$, $\mathrm{dinv}_p(\lambda)=4$

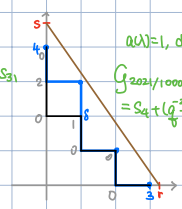
$\mathcal{G}_{2041/0000}(x;q^{-1})$

$= S_{41} + (q^{-2}+q^{-1})S_{31} + q^{-3}S_{22} + q^{-3}S_{241}$



$a(\lambda)=1$, $\mathrm{dinv}_p(\lambda)=2$

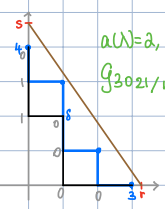
$\mathcal{G}_{3010/0001}(x;q^{-1}) = S_{41} + q^{-1}S_{31}$



$a(\lambda)=1$, $\mathrm{dinv}_p(\lambda)=3$

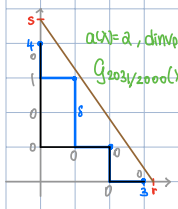
$\mathcal{G}_{2021/1000}(x;q^{-1})$

$= S_{41} + (q^{-2}+q^{-1})S_{31} + q^{-3}S_{22} + q^{-3}S_{241}$



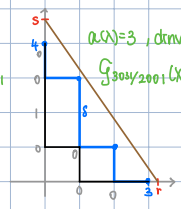
$a(\lambda)=2$, $\mathrm{dinv}_p(\lambda)=2$

$\mathcal{G}_{3021/1001}(x;q^{-1}) = S_{41} + q^{-1}S_{31} + q^{-2}S_{22}$



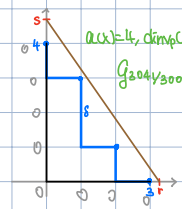
$a(\lambda)=2$, $\mathrm{dinv}_p(\lambda)=1$

$\mathcal{G}_{2031/2000}(x;q^{-1}) = S_{41} + q^{-1}S_{31}$



$a(\lambda)=3$, $\mathrm{dinv}_p(\lambda)=1$

$\mathcal{G}_{3041/2001}(x;q^{-1}) = S_{41} + q^{-1}S_{31}$



$a(\lambda)=4$, $\mathrm{dinv}_p(\lambda)=0$

$\mathcal{G}_{2041/3001}(x;q^{-1}) = S_{41}$

Def: For $\vec{b}\in\mathbb{Z}^\ell$, the generalized q,t-Catalan number $C_{\vec{b}}(q,t) :=$ coeff of $s_{|\vec{b}|}(X)$ in $\omega(D_{\vec{b}}\cdot 1)$.

(When $\vec{b}=(1,\dots,1)$, $C_{\vec{b}}(q,t)$ is the q,t-Catalan number $\langle s_\ell(X)\rangle\omega(D_{1,\dots,1}\cdot 1)$ introduced by Garsia and Mark Haiman

Corollary 5.5.5: With $\vec{b}=(b_1,\dots,b_\ell)$, r,s and $p=\frac{s}{r}$ as in Theorem 5.5.1, $C_{\vec{b}}(q,t) = \sum_\lambda t^{a(\lambda)} q^{\mathrm{dinv}_p(\lambda)}$ where the sum is over lattice paths λ from $(0,|s|)$ to $(|r|,0)$ lying weakly below the line through $(0,s)$ and $(r,0)$.